

The Hamiltonian

$$H = \sum_{i=1}^n \dot{x}_i p_i - \mathcal{L} \Rightarrow H = KE + PE$$

$$\dot{x}p \Rightarrow 2KE$$

$$|| PE(x) = cx$$

$$H = \frac{1}{2}mv^2 + cx$$

$$H = \frac{1}{2}m\dot{x}^2 + cx$$

$$H = \frac{m^2\dot{x}^2}{2m} + cx$$

$$H = \frac{p^2}{2m} + cx \Rightarrow H(p, x)$$

$$\frac{\partial H}{\partial p} = \frac{p}{m} = \frac{m}{m}v = \dot{x}$$

$$\frac{\partial H}{\partial x} = c$$

$$\frac{\partial H}{\partial p} = \dot{x}$$

$$\frac{d}{dt}\left(\frac{\partial H}{\partial p}\right) = \ddot{x}$$

$$m \frac{d}{dt} \left(\frac{\partial H}{\partial p} \right) = m \ddot{x}$$

$$m \frac{d}{dt} \left(\frac{\partial H}{\partial p} \right) + \frac{\partial H}{\partial x} = 0$$

$$\leadsto m \ddot{x} + c = 0$$

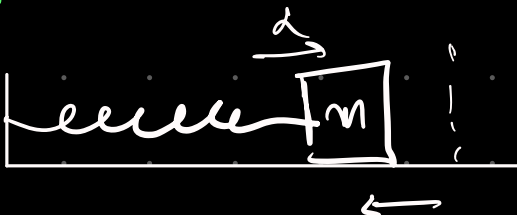
$$m \ddot{x} = -c$$

$$\ddot{x} = -\frac{c}{m}$$

$$m \frac{d}{dt} \left(\frac{\partial H}{\partial p} \right) = - \frac{\partial H}{\partial x}$$

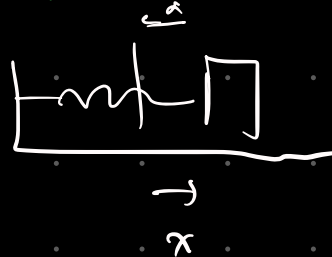
$$m a = F$$

Simple mass Spring



System

Simple harmonic
OS etc.



$$H = \frac{1}{2} m v^2 + \frac{1}{2} k x^2$$

$$\frac{\partial H}{\partial x} = kx$$

$$H = \frac{P^2}{2m} + \frac{1}{2} k x^2$$

$$\frac{\partial H}{\partial P} = \frac{P}{m} = \dot{x}$$

$$m \frac{d}{dt} \left(\frac{\partial H}{\partial p} \right) + \frac{\partial H}{\partial x} = 0$$

$$\frac{d}{dt} \frac{\partial H}{\partial p} = \ddot{x}$$

$$m \ddot{x} + kx = 0$$

$$\boxed{m \ddot{x} = -kx} \text{ equation of motion}$$

$$H = \sum_i \dot{q}_i p_i - L$$

for familiar $q = x$

for 1 generalized coordinate

$$H = \dot{x}p - L = \dot{x}p - [K E - P E]$$

$$v p - [K E - P E]$$

$$H = K E + P E$$